Radar-based Sentiment Analysis and Recognition

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Abstract

1 Introduction

Sentiment analysis commonly refers to the process of identifying and extracting subjective information from text, speech, or other forms of data [Ain *et al.*, 2017]. In this study, we focused on sentiment analysis using radar-based data. Radar sensors are a powerful tool for capturing subtle movements and gestures. They are non-intrusive, privacy-preserving, and much easier to deploy in real-world scenarios compared to other modalities such as cameras or wearable sensors [Sun *et al.*, 2020]. The data from these sensors have proven useful in applications such as gesture recognition [Sun *et al.*, 2020], object detection [Manjunath *et al.*, 2018] and human activity recognition [Froehlich *et al.*, 2024].

The main focus of researchers in the field of sentiment analysis has been on text-based and speech-based data, with a significant amount of work done on NLP models [Rokade *et al.*, 2025] and speech processing techniques. The objective was always to create models that understand human emotions through media but rarely through their body movements. Consequently, the use of radar-based data for sentiment analysis and recognition is still a relatively unexplored area, with limited research and few datasets available. The characteristics of radar signals, such as their ability to capture micro-movements and gestures, make them a promising source of information for micro-movement based experiments such as sentiment analysis [Sun *et al.*, 2020].

The objective of this paper is to explore the potential of radar-based data for sentiment analysis and recognition. We propose different approaches to transform radar signals into training data and evaluate how effective the classification models are in recognizing different sentiments. We split the paper into three different sections, each focusing on a different aspect of the problem and using a different methodology. The first focuses on feature-based classification using a mix of kinematic, behavioral and spatial features extracted from the radar signals, the second method is based on deep learning with raw sensor data, and the third relies on neural network-based interpolation and classification.

2 Related Work

This research is based on the experiments conducted by [Zuo *et al.*, 2025], in which they introduced "a multimodal radio frequency dataset" for both human behavior recognition and emotion analysis. By conducting a series of experiments on 44 participants, they collected data from 13 radars, 8 RFID tags, LoRa devices and infrared cameras in 4 different campaigns that each focused on a different aspect of human behavior. We will be interested in this paper on identifying and classifying the sentiments in the third campaign that was built around 6 sentiment states: *depression, distraction, excitement, focus, relaxation and stress*.

With the nature of the data collected, multiple approaches can be used to analyze it. The initial approach to handle radar data is to extract features from the radar signals and use them as input to classification models. This approach has been used in previous studies for human activity recognition [Deng *et al.*, 2018] and gesture recognition [Sun *et al.*, 2020]. The features extracted from the radar signals can capture the kinematic, behavioral and spatial characteristics of the target's movement, which can be useful for sentiment analysis. Another approach is the application of deep learning. Recent self-supervised learning methods have shown that powerful representations can be learned directly from raw data without manual feature engineering. SimCLR [Chen *et al.*, 2020] learns embeddings by maximizing agreement between augmented views of the same sample, producing representations that transfer well to downstream classification tasks. CLIP [Radford *et al.*, 2021] extends this idea by aligning embeddings from different modalities into a shared semantic space using contrastive learning. In the language domain, transformer architectures [Vaswani *et al.*, 2017] have demonstrated the effectiveness of self-attention for modeling complex relationships within sequential data. Motivated by these advances, our method applies transformer-based encoders directly to raw radar and infrared camera data, allowing spatial and temporal patterns to be learned automatically rather than relying on handcrafted features. The final approach is [INCLUDE THIRD METHOD BACKGROUND].

3 Dataset

As previously mentioned, the focus point of this paper is the third campaign of the RF-Behavior dataset and more specif-

ically the radar data. It is composed of a sequence of XYZ coordinates representing the position of a target in 3D space over time. Each method will use a different approach to transform the data into a format suitable for training classification models. However, before any transformation can be applied, the data must first be synchronized across different radar sensors to ensure that the data from different sensors is aligned in time, as this will allow for accurate feature extraction. Once the data is synchronized, the data is normalized on a user basis so the user-specific biases are removed. After these preprocessing steps, the data is ready to be transformed into a format suitable for training classification models.

Even when the radar data is the main concern, the infrared data can be used to provide additional information about the target's movement and behavior. The infrared data is composed of a sequence of 3 or 6 different 7D landmarks (3 position coordinates and 4 orientation quaternion) which track the target's body parts over time at a sampling rate of 100 Hz. This data can localize the bodyparts with high precision and will mainly be utilized in the second method that is based on deep learning with raw sensor data.

Table 1: Extracted trajectory features.

Feature	Description
Path Efficiency	Displacement/trajectory length
Mean Speed	Average speed
Std. Speed	Speed variability
Median Speed	Median speed
25th/75th Speed	Speed quartiles
Mean Acceleration	Average acceleration
Std. Acceleration	Acceleration variability
Median Acceleration	Median acceleration
25th/75th Acceleration	Acceleration quartiles
Mean Jerk	Average jerk
Std. Jerk	Jerk variability
Median Jerk	Median jerk
25th/75th Jerk	Jerk quartiles
Spectral Centroid	Spectrum center of mass
Spectral Entropy	Spectrum complexity
Speed CV	Normalized speed variability
Direction Change Rate	Rate of direction changes
Speed Autocorrelation	Temporal speed correlation
Tortuosity	Path curvature
Speed Skewness	Speed distribution asymmetry
Fraction Stationary Time	Stationary time proportion
Mean Turn Angle	Average turning angle
Turn Angle Entropy	Turning angle diversity
Explained Variance Ratio 1–3	PCA variance ratios

4 Method 1: Feature-Based Classification

4.1 Preprocessing

For the preprocessing step of this method, we can proceed to segment the data into smaller windows. This is done to capture the temporal dynamics of the target's movement. The window size and stride are hyperparameters that can be tuned based on the specific application and dataset. In this study, we experimented with different window sizes and strides to find the optimal configuration for our classification task. Lower window sizes tend to underperform and higher window sizes tend to overfit the data and lose generalization. The optimal window size was found to be 90 frames with a stride of 20 frames. This configuration allows for capturing the temporal dynamics of the target's movement while also providing enough data for training the classification models.

4.2 Feature Extraction

Given the preprocessed data, we can extract a set of features that capture the kinematic, behavioral and spatial characteristics of the target's movement. The features are extracted from each window of data and are used as input to the classification models. The features are divided into three categories: motion dynamics, statistical and directional features. The motion dynamics features capture the speed, acceleration and jerk of the target's movement. The statistical features capture the distribution of the target's movement in terms of speed, acceleration and jerk. The directional features capture the directionality of the target's movement in terms of turn angles and explained variance ratios. The extracted features are summarized in Table 1.

4.3 Training Loop

With the current setup, the data is now composed of 27 features per radar sensor. Given that we have 13 radar sensors, the total number of features is 351 over more than 600 windows. For an effective training of the classification models, we need to reduce the dimensionality of the data and select the most relevant features. To find the optimal configuration of window size, stride and number of features, we implemented a training loop that iterates over different configurations and evaluates the performance of the classification models using Leave-One-Subject-Out (LOSO) cross-validation. It uses the ANOVA F-test to select the top-k features for each radar sensor and concatenates them to form the final feature set. ANOVA F-test is a statistical method used to determine the significance of differences between group means [Perangin-Angin and Bachtiar, 2021], it is widely used in feature selection tasks where the goal is to identify the most informative features for classification. The classification models are then trained on the training set and evaluated on the test set. The accuracy and weighted F1-score are recorded for each configuration and the best configuration is selected based on the highest mean accuracy across all folds. The training loop is summarized in Algorithm 1.

4.4 Model selection

The classification models considered in this study include Support Vector Machine (SVM), Gradient Boosting (LightGBM, XGBoost, CatBoost) and Multi-Layer Perceptron (MLP). SVM is a popular classification model that finds the optimal hyperplane that separates the data into different classes [Kazemi *et al.*, 2022]. Gradient Boosting is an ensemble learning method that combines multiple weak classifiers to create a strong classifier [Ryan and Massaron, 2025]. MLP is a type of neural network that consists of multiple layers of interconnected nodes [Alsmadi *et al.*, 2009]. The hyperparameters of each model are tuned using grid search and the best configuration is selected based on the highest mean accuracy across all folds.

4.5 Results

Figure 1 compares the performance of the evaluated classifiers using the mean F1-score across all folds. Among

Algorithm 1 Feature Selection and LOSO Classification

for each window configuration (w, s) **do**
  for each feature count $k \in K$ **do**
    for each LOSO split **do**
      Split data into training and testing sets.
      for each radar **do**
        Select top-k features (ANOVA F-test).
      end for
      Concatenate selected features.
      for each classifier $m \in \mathcal{M}$ **do**
        Train, evaluate, and record accuracy and weighted F1-score.
      end for
    end for
    Compute mean metrics across LOSO folds.
  end for
end for
Select the best (w, s, k) for each classifier.
Output: Optimal configuration and corresponding accuracy and weighted F1-score.

the considered models, the Support Vector Machine (SVM) achieved the highest performance with a mean F1-score of 0.72 which suggests that it was able to identify robust decision boundaries.

The gradient boosting models also achieved competitive results. Both LightGBM and XGBoost obtained mean F1-scores between 0.66 and 0.67, indicating that ensemble tree-based methods are capable of capturing the non-linear relationships present in the extracted features. Their performance, while slightly lower than that of the SVM, confirm that boosting-based approaches can be effective for radar-based sentiment recognition tasks. However, CatBoost underperformed compared to the other gradient boosting models, achieving a mean F1-score of 0.62. While it is known for its strong performance on heterogeneous tabular datasets [Ryan and Massaron, 2025], the specific characteristics of the radar-based features may have limited its effectiveness in this context.

The Multi-Layer Perceptron (MLP) achieved a similar performance to the gradient boosting models, with a mean F1-score in the same range. Although neural networks are generally effective at learning complex feature interactions, the relatively limited size of the dataset and the use of engineered features may have limited the advantages typically associated with deep learning approaches.

Overall, the superior performance of the SVM highlights the effectiveness of the proposed feature extraction pipeline and suggests that carefully engineered motion descriptors can provide sufficient discriminative power without requiring more complex deep learning architectures.

5 Method 2: Deep Learning with Raw Data

5.1 Dataset

For this method, we utilize the radar and infrared camera from the first three campaigns of the RF-Behavior dataset. Behaviors with highly repetitive actions (e.g., walking and running

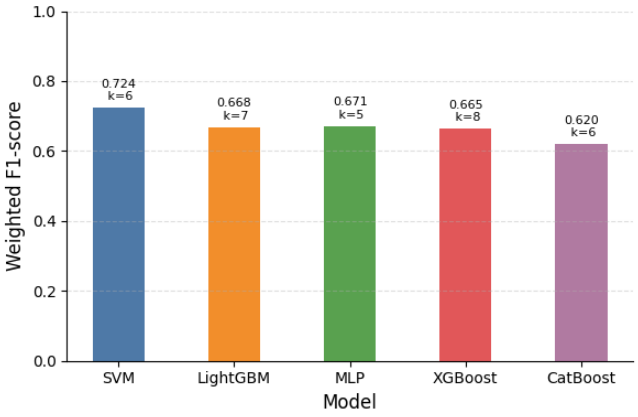


Figure 1: Models comparison based on F1 score.

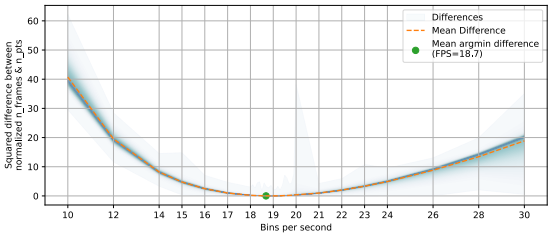


Figure 2: Optimization curve showing how we balance the number of radar frames and spatial points per frame.

from campaign 2, and the six emotions from campaign 3) are segmented into random intervals of 4 to 6 seconds to match the typical duration of other activities. This segmentation introduces diversity and prevents the model from memorizing irrelevant long-term patterns, such as a participant's specific walking path, rather than learning the actual behavior itself.

5.2 Preprocessing: Uniform Batches

Sampling Rate

Each of the 13 radar sensors captures an irregular number of 4D points (comprising density and X, Y, Z coordinates) at arbitrary timestamps. To structure this sparse data into uniform batches, represented as (batch, time, points, coordinates), we group the data into evenly spaced time bins. We calculate the optimal frame rate empirically by testing values from 10 to 30 frames per second (FPS) at 0.1 increments across all recordings. Across all candidate FPS, we normalize the total number of frames and the average number of points per frame, and compute the squared difference between the two corresponding quantities. This optimization yields a U-shaped error curve (Figure 2) with a minimum at 18.7 FPS (representing a 53.5 ms time bin). At this rate, the average recording's frame count matches its average points-per-frame count, balancing the temporal and spatial dimensions of our input tensor.

Jagged Input Tensors

Because recording durations vary across samples, the resulting tensors are uneven (jagged) over time. We resolve this

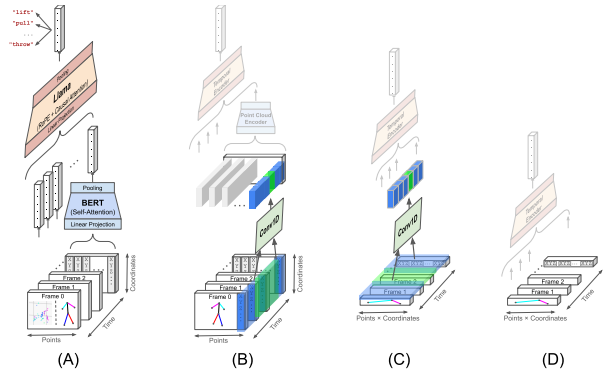


Figure 3: Deep learning pipelines. The model uses a spatial encoder to process 3D points in each frame and a temporal encoder to process sequence of latent vectors. All configurations work with the infrared camera, but only configuration (A) is compatible with radar.

by padding shorter sequences on the right using copies of the recording's final frame. In this work, we did not explore the application of other padding strategies.

Another challenge is that the number of spatial points per frame varies based on active sensors and the specific campaign. Only three upper-body landmarks are tracked via the infrared camera in campaign 1, whereas campaigns 2 and 3 add three lower-body landmarks. Furthermore, individual radar sensors often return varying amounts of data depending on the participant's distance. To handle this structural variability without locking the network into a fixed layout, we divide each batch of recordings into smaller mini-batches of individual frames that share an identical point count. These uniform subsets are processed independently through the network before being reassembled and combined into the final output tensors.

5.3 Model

Baseline Architecture: Transformers

We design a neural network architecture using two sequential Transformer stacks to condense the variable time and spatial point dimensions (see Figure 3) to size one.

To process the points dimension, we build a Point Cloud Encoder (PCE) that uses a linear projection layer followed by a BERT [Devlin *et al.*, 2019] self-attention encoder. The PCE treats the points in a single frame as a sequence of tokens, learning their spatial relationships to produce a single vector for that frame. Because spatial points have no sequential order, we omit positional encodings for radar. For the infrared camera, we can use learned positional embeddings added to the landmark projections to help the model identify specific body parts. The PCE maps inputs of shape (batch, points, coordinates) to a representation of shape (batch, hidden_size).

To process the time dimension, we implement a Temporal Encoder consisting of a linear projection layer followed by a LLaMA [Touvron *et al.*, 2023] decoder. To capture how physical gestures unfold over time, this encoder uses rotary positional embeddings and causal attention to model sequential relationships across frames. We apply a pooling operation over the output vectors to generate a single summary embedding for the entire recording. An attention mask ensures that the padding frames added during preprocessing are ignored. The temporal encoder processes inputs of shape (batch, time, hidden_size) and outputs a tensor of shape (batch, hidden_size).

Hyperparameter	Values
# Convolutional Blocks	0-4
# Convolutional Channels	2^i (i = 4, ..., 7)
# Transformer blocks	1-6
# Attention Heads	2^i (i = 2, ..., 5)
Embedding Size	2^i (i = 4, ..., 10)
Pooling Strategy	Mean, CLS, EOS
Positional Embeddings (infrared BERT)	Yes/No

Table 2: Architectural hyper-parameters evaluated during grid search.

For the infrared camera, we also design an optional 1D convolutional downsampling block with a stride of 2 to halve the temporal frame rate and improve efficiency. This block can first flatten all coordinates of all landmarks per frame into a single vector (Configuration 3C), but that requires a fixed sensor setup. Alternatively, Configuration 3B processes each landmark independently using shared weights, allowing the use of PCE to handle any number of tracking points.

Architectural Hyperparameters

To optimize this highly scalable Transformer architecture, we define a search space for its key hyper-parameters (Table 2). We calculate model sizes for 10,000 randomly sampled configurations and group them into 20 geometrically spaced bins ranging from 70k to 100M parameters. We then sequentially process one random model from each bin and evaluate its performance.

Supervised Classification

We evaluate and tune these configurations using a supervised classification task. To avoid leakage of an individual's quirks, we partition the dataset by user IDs. The test set contains four random users, the validation set contains two, and the remaining users are allocated to the training set (approximately 160 recordings per behavior class). We set output projection size of the above model to the number of behavior classes. The network is trained using cross-entropy loss, the AdamW optimizer (learning rate of 10^{-3}, batch size of 32), and early stopping with a patience of 5 epochs. We save the model with the lowest validation loss and evaluate it on the test set using a weighted F1-score. All experiments were run on an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB GDDR6 VRAM).

Unsupervised Contrastive Learning

To make the model useful beyond the specific behaviors in the RF-Behavior dataset, we evaluate its ability to learn general-purpose features without relying on rigid class labels. We map the final output to a 256-dimensional space and train the network in a semi-supervised manner using the NT-Xent loss from SimCLR [Chen *et al.*, 2020] with the rest of the training parameters held the same as the supervised approach. This loss function pulls the embedding vectors of different recordings of the same behavior closer together in vector space,

Campaign	# Classes	Modality	# Params	Test F1	# Models
C1	21	Radar	1,311,125	94.41%	12
C1	21	Infrared	526,869	96.56%	25
C2	10	Radar	106,826	91.97%	23
C2	10	Infrared	1,541,850	98.12%	29
C3	6	Radar	1,349,766	46.12%	30
C3	6	Infrared	243,798	56.16%	56
C1+C2	31	Radar	120,895	89.51%	6
C1+C2	31	Infrared	195,375	91.52%	5
C1+C2+C3	37	Radar	1,738,021	71.76%	1
C1+C2+C3	37	Infrared	165,269	78.29%	7

Table 3: Grid search: Performance metrics of supervised classification models.

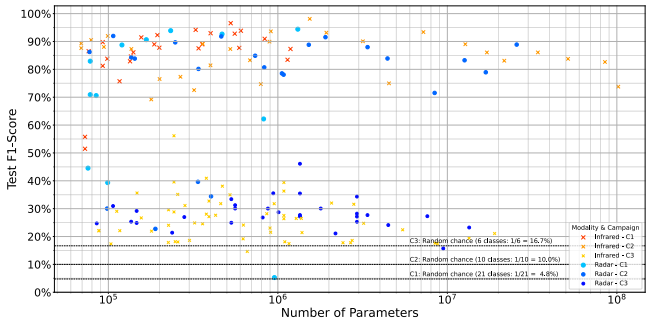


Figure 4: F1-score distribution of our supervised models. Each point represents a unique model configuration evaluated during our hyperparameter search.

while pushing different behaviors far apart. We evaluate this approach using 1-shot classification on two unseen test behaviors, holding out another two for validation. The NT-Xent loss for a positive pair (i, j) and temperature τ is defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2N} 1_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}, \quad (1)$$

Because one of the modalities will be more difficult to model than the other, we align the embedding spaces of both modalities so the weaker sensor can learn from the stronger one. We project both types of sensor data into vectors of the same size and with a symmetric contrastive loss, similar to CLIP [Radford *et al.*, 2021], we train both models jointly. Using cross entropy, this loss function maximizes the similarity between simultaneous radar and infrared recordings of the same event while minimizing similarity across different events.

5.4 Results

Our hyperparameter search revealed that using classification token (CLS) pooling in the PCE collapses the model's output space, mapping all inputs to nearly identical vectors. Conversely, using convolutional blocks to downsample the temporal dimension in the infrared camera model drastically reduces training time, as the self-attention mechanism's computational cost scales quadratically with sequence length. For radar models, a larger PCE increases inference time due to our mini-batched-frames by mini-batched-frames processing of jagged points dimension. For unsupervised models, mean pooling consistently out-performed end-of-sequence (EOS) pooling. We also observed that, for unsupervised models, minimizing training loss, rather than stopping prematurely solely based on validation loss, is critical for test performance, as overly early termination leads to noisy, unconverged models.

Supervised results are summarized in Table 3, with the overall F1-score distribution plotted in Figure 4.

For the challenging emotion classes in campaign 3, we attempted to improve accuracy by using a genetic algorithm to mutate and combine the top-performing architectures. However, this did not yield better models within our computational limits. We expect this likely stems from the subtle, microscopic nature of emotional expressions, which may require denser sensor arrays or more tracking landmarks to capture. Across all campaigns, the radar performed worse than the infrared camera, which we attribute to the radar's lower spatial resolution, sparser data collection, and higher signal noise.

	# Behaviors	Modality	# Params	Test F1	# Models
C1	17-2-2	Radar	254,144	93.03%	6
C1	17-2-2	Infrared	142,016	96.68%	7
C2	6-2-2	Radar	93,440	77.51%	14
C2	6-2-2	Infrared	923,280	93.48%	31

Table 4: Grid search: One-shot classification results for unsupervised models. The behaviors column indicates the split of train, validation, and test classes.

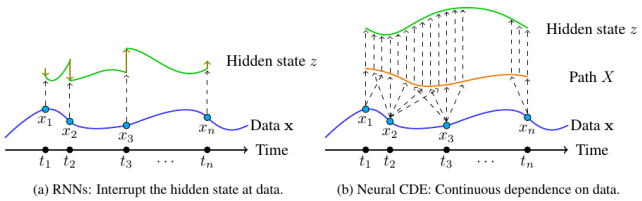
Our unsupervised models were evaluated using 1-shot classification, where we measure cosine similarity for each query recording with one reference recording per unseen class. This evaluation was repeated using cross-validation across all samples. The results (Table 4) show that the unsupervised models can generalize remarkably well to entirely new behaviors, even matching supervised performance while requiring only a single reference example and significantly fewer parameters.

6 Method 3: Neural Network-Based Interpolation and Classification

The radar-based dataset consists of 13 separate radar sensors each recording data at different timestamps, in order to unify the sensors to a single timeline, interpolation is required. This method consists of performing interpolation using Neural Networks and comparing with classical methods, emotion classification is also performed.

6.1 Preprocessing

Using the preprocessing sample code provided in the dataset, the radar sensors' data were converted to 39 XYZ coordinates, afterwards, the timestamps were rounded to 1 decimal place and the sensors concatenated together into a single csv file for each user for each emotion. These were then concatenated together to get a singular csv file containing the processed data of every user, emotion and sensor. Grouping by user_id and emotion, a sparse sequence of observations from all the sensors combined can be retrieved.



(a) RNNs: Interrupt the hidden state at data. (b) Neural CDE: Continuous dependence on data.

Figure 5: Comparison of hidden state updates between RNN and NCDE.

To avoid any overlap between dataset splits and to test on unseen users, the user_ids were split into train, valid and test sets with ratios 0.6/0.2/0.2 . Normalization is performed by setting mean to 0 to allow the model to only train on the residual. Min-max normalization is applied to the timestamps to set the minimum to 0 and maximum to 1.

While training the models, 0.1 to 0.3 of the sequence is randomly set aside as the target sequence, this is set to 0.2 for validation and testing with a static random state for reproducibility. An observation mask containing the non-Nan values is obtained for both the input and target sequence, this is combined with the input sequence to allow the model to focus on observations and is used to train the model using only non-Nan target values. Last Observation Carried Forward (LOCF) is then applied on the data to counter-act the sparsity of the dataset.

6.2 Model Selection

The proposed method consists of training a Neural Controlled Differential Equation (NCDE) model by Kidger et. al [Kidger *et al.*, 2020] for interpolation and classification using the torchcde library. It was selected due to it's ability to deal with irregular timestamps. As shown in Figure. 5, the main advantage over Recurrent Neural Networks (RNNs) is the continuous update of it's hidden state, this is achieved by solving and integrating the differential equation in Equation. 2 where $f(t, z(t))$ and $g(x_0)$ are Neual Networks (NNs) and X is a Hermite cubic splines interpolation of sparse input sequence.

$$\frac{dz(t)}{dt} = f(t, z(t))\frac{dX(t)}{dt}, \qquad z(t_0) = g(x_0), \qquad (2)$$

Shown in Figure. 6 is the workflow of the proposed method, it starts with interpolating the input sequence using a differentiable hermite cubic spline to obtain a continuous control data X, whose first instance x(0) is passed through an MLP to get z(0) and both are fed into the cdeint function along with the target timestamps dictating at what time we want the hidden states z(t), the output sequence is fed into an MLP decoder to get the predicted values while the final hidden state is fed into an MLP classifier to get emotion probabilities.

6.3 Metrics

The model was trained using a combination of Mean Squared Error (MSE) loss for the sensor interpolation and Cross-Entropy loss for the emotion classification. Accuracy was used to measure the classification performance of the model on the test set.

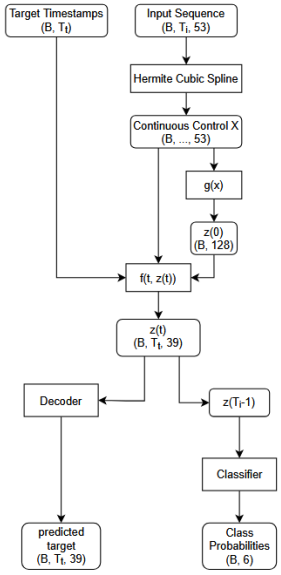


Figure 6: Workflow of the proposed method. It uses a differentiable Hermite cubic spline that can be integrated to obtain the hidden state at arbitrary timestamps. The final hidden state is used to obtain class probabilities, similar to RNNs.

Method	**MSE**	**Cross-Entropy**	**Accuracy**
NCDE model (final epoch)	0.0992	1.7809	30.55
NCDE model (best classify loss)	0.0992	1.7809	30.55
NCDE model (best MSE loss)	0.0999	1.7913	22.22

Table 5: NCDE model performance on test set.

6.4 Results

The model was trained for 20 epochs and the weights with the best MSE loss and cross-entropy loss on valid set were saved along with final epoch weights. The performance of the model on the test set is listed in Table. 5. Despite the small MSE loss, the model still performs poorly when interpolating, this is due to the mean of the input sequence being 0, leading to a very small loss and the model output being close to 0. The same effect can be gotten by just taking the mean of the input sequence as the predicted value.

Further comparison between regular cubic spline and NCDE is required to determine if the model is underfitting or if the input sequences does not have enough discriminatory information.

6.5 Discussion

A key limitation of our deep learning approach is that our Transformer encoders were adapted from general-purpose language models. We did not explore specialized point-cloud architectures or lighter temporal designs, nor did we evaluate

robustness to unseen physical sensor layouts or run thorough training method ablation studies.

To improve future generalization, several data augmentation strategies could be introduced during training: 1) randomly dropping active sensors out of the 13 radars, 2) downsampling high-density radar frames, 3) randomly dropping temporal frames, 4) cropping recordings to arbitrary durations, 5) injecting Gaussian noise into spatial coordinates, and 6) applying geometric transformations such as 3D rotations, scaling, and translations.

Additionally, our deep learning modules are compatible with all raw data formats in the RF-Behavior dataset. A unified model could be constructed using a single shared temporal head paired with sensor-specific input blocks (such as downsamplers or point cloud encoders). This would allow a single network to be trained on all sensor types and behaviors simultaneously. Finally, aligning this sensor embedding space with a pre-trained text encoder would unlock zero-shot classification, potentially allowing the model to recognize brand-new behaviors based purely on written descriptions.

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